

Catalyst::

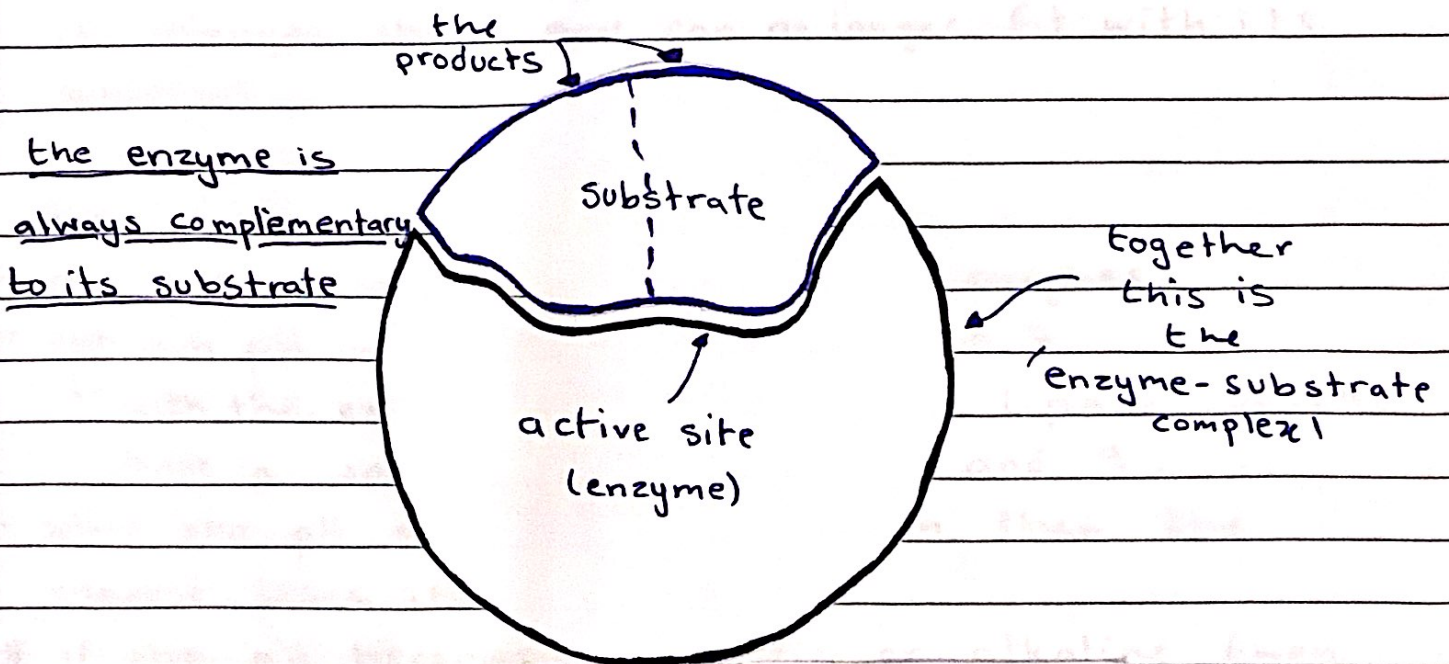
a substance that increases the rate of a chemical reaction and is not changed by the reaction.

Enzymes::

a protein that is involved in all metabolic reactions, where it functions as a biological catalyst.

enzymes in living organisms::

(in digestion) enzymes increase the reaction rate so the food is digested quicker and easier.



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effect of temperature and pH on enzymes:

temperature :-

- high temperatures are damaging for enzymes
- optimum temperature for human enzymes is 40°C
- when the temperature exceeds 40°C , the enzyme molecules start to lose their shape.
- * the process which 'modifies' the molecular structure of a protein by heat is called 'denaturation'.
- * as temperature increases, the chance of substrate molecules and enzymes colliding increases, this continues to an optimum temperature.
- * when the bonds between the enzyme start to break, it changes shape and can no longer fit with its substrate.

pH:-

- very high or low pH is damaging for enzymes
- optimum pH for human enzymes is pH 7
 - ↳ with the exceptions of protease or lipase which have a optimum pH level of 2 and 9.
- when the pH goes too up or down then the enzyme loses its shape.
- * if the pH becomes too acidic or alkaline then they are denatured.